

2019년 1학기 중간고사 문제 풀이

1. (1) $f'(x) = (3x^2 + 1)\cosh(x^3 + x - 2) > 0$ 이므로, $f(x)$ 는 증가함수이고, 역함수 f^{-1} 가 존재한다.

$$(2) L(x) = (f^{-1})'(3)(x-3) + (f^{-1})(3)$$

$$= \frac{1}{f'(1)}(x-3) + (f^{-1})(3)$$

$$= \frac{1}{4}(x-3) + 1$$

$$(3) f^{-1}(f(x)) = x \rightarrow (f^{-1})''(f(x)) = -\frac{(f^{-1})'(f(x))f''(x)}{(f'(x))^2}$$

$$f''(x) = (3x^2 + 1)^2 \sinh(x^3 + x - 2) + 6x \cosh(x^3 + x - 2)$$

$$\therefore f''(1) = 6, \quad (f^{-1})''(3) = -\frac{3}{32}$$

$$2. (1) 0 < x < 1, \quad y = \operatorname{sech}^{-1} x \Leftrightarrow x = \operatorname{sech} y = \frac{2}{e^y + e^{-y}}$$

$$\Rightarrow xe^{2y} - 2e^y + x = 0$$

$$\Rightarrow e^y = \frac{1 \pm \sqrt{1-x^2}}{x}, \quad 0 < x < 1, \quad y > 0, \quad e^y > 1 \text{ 이므로}$$

$$\Rightarrow e^y = \frac{1 + \sqrt{1-x^2}}{x}$$

$$\text{그러므로 } y = \ln\left(\frac{1 + \sqrt{1-x^2}}{x}\right), \quad 0 < x < 1 \text{ 이다.}$$

$$(2) 0 < x < 1, \quad y = \operatorname{sech}^{-1} x \Leftrightarrow x = \operatorname{sech} y$$

$$\Rightarrow 1 = -\operatorname{sech} y \tanh y \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} = -\frac{1}{\operatorname{sech} y \tanh y}.$$

한편 $1 - \tanh^2 y = \operatorname{sech}^2 y$ 이고 $0 < x < 1$ 에서 $y > 0$ 이다. 이때 $\tanh y > 0$ 이므로 $\tanh y = \sqrt{1-x^2}$ 이고,

$$\text{따라서 } \frac{dy}{dx} = \frac{-1}{x\sqrt{1-x^2}} \text{ 이다.}$$

$$(3) \left(\tan^{-1}\left(\frac{x}{y}\right) - 2\right)^3 - y^3 = 2 \sin^{-1}(xy) - 9 \text{ 에서 } x=0, \quad y=1 \text{ 이다. 음함수 미분법에 의해}$$

$$3\left(\tan^{-1}\left(\frac{x}{y}\right) - 2\right)^2 \left(\frac{\frac{y-xy'}{y^2}}{1+\left(\frac{x}{y}\right)^2}\right) - 3y^2 y' = 2 \frac{(y+xy')}{\sqrt{1-(xy)^2}}$$

이다. 따라서 $x=0, y=1$ 에서 $y' = \frac{10}{3}$ 이므로 접선의 방정식은 $y = \frac{10}{3}x + 1$ 이다.

$$3. (1) \lim_{x \rightarrow 0^+} x \ln x : 0 \times \infty \text{ 꼴}$$

$$= \lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x}} : \frac{\infty}{\infty} \text{ 꼴, 로피탈 정리 이용}$$

$$= \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{\frac{1}{x^2}} = 0$$

$$(2) \lim_{x \rightarrow 0^+} \left(\frac{1}{x} - \frac{1}{\sin^{-1} x} \right) : (\infty - \infty) \text{ 꼴}$$

$$= \lim_{x \rightarrow 0^+} \frac{\sin^{-1} x - x}{x \sin^{-1} x} : \left(\frac{0}{0} \right) \text{ 꼴, 로피탈 정리 이용}$$

$$= \lim_{x \rightarrow 0^+} \frac{1 - \sqrt{1-x^2}}{\sqrt{1-x^2} \sin^{-1} x + x} : \left(\frac{0}{0} \right) \text{ 꼴, 로피탈 정리 이용}$$

$$= \lim_{x \rightarrow 0^+} \frac{x}{x \sin^{-1} x + 2\sqrt{1-x^2}} = 0$$

$$(3) \lim_{x \rightarrow \infty} \frac{\ln \left(\int_0^x e^t dt \right)}{x^2} = \lim_{x \rightarrow \infty} \frac{e^{x^2}}{2x \int_0^x e^{t^2} dt} \quad (\because \text{로피탈법칙 적용})$$

$$= \lim_{x \rightarrow \infty} \frac{2xe^{x^2}}{2 \int_0^x e^{t^2} dt + 2xe^{x^2}} \quad (\because \text{로피탈법칙 적용})$$

$$= \lim_{x \rightarrow \infty} \frac{2e^{x^2} + 4x^2 e^{x^2}}{2e^{x^2} + 2e^{x^2} + 4x^2 e^{x^2}} = 1$$

$$\therefore \lim_{x \rightarrow \infty} \left(\int_0^x e^{t^2} dt \right)^{\frac{1}{x^2}} = \lim_{x \rightarrow \infty} e^{\frac{1}{x^2} \ln \left(\int_0^x e^{t^2} dt \right)} = e^{\lim_{x \rightarrow \infty} \frac{1}{x^2} \ln \left(\int_0^x e^{t^2} dt \right)} = e$$

$$4. (1) x_n = \frac{1}{n} \sqrt[n]{(n+1)(n+2) \cdots (n+n)} \text{ 이므로,}$$

$$\begin{aligned} \ln x_n &= \ln \frac{\sqrt[n]{(n+1)(n+2) \cdots (n+n)}}{n} = \ln \sqrt[n]{\left(1 + \frac{1}{n}\right) \left(1 + \frac{2}{n}\right) \cdots \left(1 + \frac{n}{n}\right)} \\ &= \frac{1}{n} \left\{ \ln \left(1 + \frac{1}{n}\right) + \cdots + \ln \left(1 + \frac{n}{n}\right) \right\} = \frac{1}{n} \sum_{k=1}^n \ln \left(1 + \frac{k}{n}\right) \\ &= \int_0^1 \ln(1+x) dx = [x \ln(1+x)]_0^1 - \int_0^1 \frac{x}{1+x} dx \\ &= 2\ln 2 - 1 = \ln \left(\frac{4}{e} \right). \end{aligned}$$

$$\therefore \lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} e^{\ln x_n} = e^{\frac{\ln 4}{e}} = \frac{4}{e}.$$

$$\begin{aligned} (2) \int x^3 (\ln x)^3 dx &= \frac{1}{4} x^4 (\ln x)^3 - \frac{3}{4} \int x^3 (\ln x)^2 dx \quad (\because \text{부분적분 이용}) \\ &= \frac{1}{4} x^4 (\ln x)^3 - \frac{3}{4} \left(\frac{1}{4} x^4 (\ln x)^2 - \frac{1}{2} \int x^3 \ln x dx \right) \quad (\because \text{부분적분}) \\ &= \frac{1}{4} x^4 (\ln x)^3 - \frac{3}{4} \left(\frac{1}{4} x^4 (\ln x)^2 - \frac{1}{2} \left(\frac{1}{4} x^4 \ln x - \frac{1}{16} x^4 \right) \right) + C \quad (\because \text{부분적분}) \end{aligned}$$

$$\begin{aligned}
\int_0^1 x^3 (\ln x)^3 dx &= \lim_{t \rightarrow 0^+} \int_t^1 x^3 (\ln x)^3 dx \\
&= \lim_{t \rightarrow 0^+} \left[\frac{1}{4} x^4 (\ln x)^3 - \frac{3}{16} x^4 (\ln x)^2 + \frac{3}{32} x^4 \ln x - \frac{3}{128} x^4 \right]_t^1 \\
&= -\frac{3}{128}
\end{aligned}$$

5. (1) $\int \frac{x^2}{(x-3)(x-2)^2} dx = \int \frac{A}{(x-3)} + \frac{B}{(x-2)} + \frac{C}{(x-2)^2} dx$ 로부터 $A=9, B=-8, C=-4$ 이므로

$$\int \frac{x^2}{(x-3)(x-2)^2} dx = \int \frac{9}{(x-3)} + \frac{-8}{(x-2)} + \frac{-4}{(x-2)^2} dx = 9 \ln |x-3| - 8 \ln |x-2| + \frac{4}{x-2} + C \text{ 이다.}$$

$$\begin{aligned}
(2) \int_0^{\pi/2} \frac{1}{2-\cos x} dx &= \int_0^1 \frac{\frac{2}{1+t^2}}{2-\frac{1-t^2}{1+t^2}} dt = \int_0^1 \frac{2}{3t^2+1} dt = \left[\frac{2}{3} (\sqrt{3} \tan^{-1}(\sqrt{3}t)) \right]_{t=0}^{t=1} \\
&= \frac{2\sqrt{3}\pi}{9} \left(= \frac{2\pi}{3\sqrt{3}} \right)
\end{aligned}$$

(3) - Sol1) $\int \frac{x^2 \tan^{-1} x}{x^2+1} dx = \int \left(1 - \frac{1}{x^2+1} \right) \tan^{-1} x dx$

$$\begin{aligned}
&= \int \tan^{-1} x dx - \int \frac{1}{x^2+1} \tan^{-1} x dx \\
&= x \tan^{-1} x - \int \frac{x}{1+x^2} dx - \int \frac{1}{x^2+1} \tan^{-1} x dx \\
&= x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) - \frac{1}{2} (\tan^{-1} x)^2 + C, \quad C \text{ 는 상수.}
\end{aligned}$$

Sol2) $\int \frac{x^2 \tan^{-1} x}{x^2+1} dx = \int u \tan^2 u du = \int u (\sec^2 u - 1) du$, (here $\tan^{-1} x = u, \frac{1}{1+x^2} dx = du$)

$$\begin{aligned}
&= \int u \sec^2 u du - \int u du = u \tan u - \int \tan u du - \frac{1}{2} u^2 \\
&= u \tan u + \ln |\cos u| - \frac{1}{2} u^2 + C \\
&= x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) - \frac{1}{2} (\tan^{-1} x)^2 + C, \quad C \text{ 는 상수.}
\end{aligned}$$

(4) $\frac{3x}{x^3+1} = \frac{3x}{(x+1)(x^2-x+1)} = \frac{A}{x+1} + \frac{Bx+C}{x^2-x+1} = \frac{-1}{x+1} + \frac{x+1}{x^2-x+1}$ 이고,

$$\begin{aligned}
\int_0^1 \frac{3x}{x^3+1} dx &= \int_0^1 \left(\frac{-1}{x+1} + \frac{x+1}{x^2-x+1} \right) dx \\
&\quad (x^2-x+1 = (x-\frac{1}{2})^2 + \frac{3}{4} \text{ 이 용하면}) \\
&= \left[-\ln(x+1) + \frac{1}{2} \ln(x^2-x+1) + \frac{3}{2} \cdot \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2}{\sqrt{3}} \left(x - \frac{1}{2} \right) \right) \right]_0^1 \\
&= \frac{\pi}{\sqrt{3}} - \ln 2
\end{aligned}$$

6.

$$\begin{aligned}
\int_{-2}^0 2\pi(x+2)|(x+1)(2x+3)| dx &= \int_{-2}^{-1} 2\pi(x+2)(-1)(x+1)(2x+3)^2 dx + \int_{-1}^0 2\pi(x+2)(x+1)(2x+3)^2 dx \\
&= 10\pi
\end{aligned}$$

따라서 $\alpha = 10$ 이고 $15\alpha = 150$ 이다.

$$7. \int_0^{\infty} \frac{1}{x^{1/p} + x^p} dx$$

$$\textcircled{1} \quad 0 < p < 1$$

$$\text{i) } \int_0^1 \frac{1}{x^{1/p} + x^p} dx$$

$$0 < \frac{1}{x^{1/p} + x^p} < \frac{1}{x^p} \quad \text{이므로} \quad \int_0^1 \frac{1}{x^p} dx = \lim_{t \rightarrow 0+} \int_t^1 \frac{1}{x^p} dx = \lim_{t \rightarrow 0+} \left[\frac{1}{1-p} x^{1-p} \right]_t^1 \\ = \lim_{t \rightarrow 0+} \left[\frac{1}{1-p} (1 - t^{1-p}) \right] = \frac{1}{1-p} \quad : \text{ 수렴.}$$

$$\text{비교정리에 의하여 } 0 < p < 1 \text{이면, } \int_0^1 \frac{1}{x^{1/p} + x^p} dx : \text{ 수렴.}$$

$$\text{ii) } \int_1^{\infty} \frac{1}{x^{1/p} + x^p} dx$$

$$0 < \frac{1}{x^{1/p} + x^p} < \frac{1}{x^{1/p}} \quad \text{이므로} \quad \int_1^{\infty} \frac{1}{x^{1/p}} dx : \text{ 수렴 } (\because \frac{1}{p} > 1).$$

$$(\text{또는 } \int_1^{\infty} \frac{1}{x^{1/p}} dx = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x^{1/p}} dx = \lim_{t \rightarrow \infty} \left[\frac{1}{1-1/p} x^{1-1/p} \right]_1^t = \frac{p}{1-p} : \text{ 수렴.})$$

$$\therefore \text{i) ii)에 의해 } 0 < p < 1 \text{ 일 때 } \int_0^{\infty} \frac{1}{x^{1/p} + x^p} dx \text{ 은 수렴한다.}$$

$$\textcircled{2} \quad p = 1$$

$$\int_0^{\infty} \frac{1}{2x} dx \quad \text{에서} \quad \int_1^{\infty} \frac{1}{2x} dx = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{2x} dx = \frac{1}{2} \lim_{t \rightarrow \infty} [\ln x]_1^t = \frac{1}{2} \lim_{t \rightarrow \infty} \ln t = \infty : \text{ 발산.}$$

$$\textcircled{3} \quad p > 1$$

$$\text{i) } \int_0^1 \frac{1}{x^{1/p} + x^p} dx$$

$$0 < \frac{1}{x^{1/p} + x^p} < \frac{1}{x^{1/p}} \quad \text{이므로}$$

$$\int_0^1 \frac{1}{x^{1/p}} dx = \lim_{t \rightarrow 0+} \int_t^1 \frac{1}{x^{1/p}} dx = \lim_{t \rightarrow 0+} \left[\frac{1}{1-1/p} x^{1-1/p} \right]_t^1 = \frac{p}{p-1} : \text{ 수렴.}$$

$$\text{ii) } \int_1^{\infty} \frac{1}{x^{1/p} + x^p} dx$$

$$0 < \frac{1}{x^{1/p} + x^p} < \frac{1}{x^p} \quad \text{이므로} \quad \int_1^{\infty} \frac{1}{x^p} dx : \text{ 수렴 } (\because p > 1).$$

$$(\text{또는 } \int_1^{\infty} \frac{1}{x^p} dx = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x^p} dx = \lim_{t \rightarrow \infty} \left[\frac{1}{1-p} x^{1-p} \right]_1^t = \frac{1}{p-1} : \text{ 수렴.})$$

$$\therefore \text{i) ii)에 의해 } 1 < p < \infty \text{ 일 때 } \int_0^{\infty} \frac{1}{x^{1/p} + x^p} dx \text{ 은 수렴한다.}$$

$$\textcircled{1}, \textcircled{2}, \textcircled{3} \text{에 의해 } 0 < p < 1 \text{ 이거나 } p > 1 \text{ 이면 특이 적분은 수렴한다.}$$

$$8. \quad S = \int 2\pi y ds = \int_0^2 2\pi e^x \sqrt{1+e^{2x}} dx = \int_1^{e^2} 2\pi y \sqrt{1+\frac{1}{y^2}} dy = \int_1^{e^2} 2\pi \sqrt{1+y^2} dy \quad \text{이므로,}$$

$$S = \int_1^{e^2} 2\pi \sqrt{1+y^2} dy = \int_{\frac{\pi}{4}}^{\tan^{-1} e^2} 2\pi \sec \theta \cdot \sec^2 \theta d\theta \quad (y = \tan \theta, dy = \sec^2 \theta d\theta \text{ 이요})$$

$$= 2\pi \int_{\frac{\pi}{4}}^{\alpha} \sec^3 \theta d\theta \quad (\alpha = \tan^{-1} e^2) = 2\pi \cdot \frac{1}{2} [\sec \alpha \tan \alpha + \ln |\sec \alpha + \tan \alpha|]_{\frac{\pi}{4}}^{\alpha}$$

$$= \pi (\sqrt{1+e^4} e^2 + \ln(\sqrt{1+e^4} + e^2) - \sqrt{2} - \ln(\sqrt{2}+1)) \quad (\alpha = \tan^{-1} e^2 \Rightarrow \tan \alpha = e^2 \text{ 이요}).$$